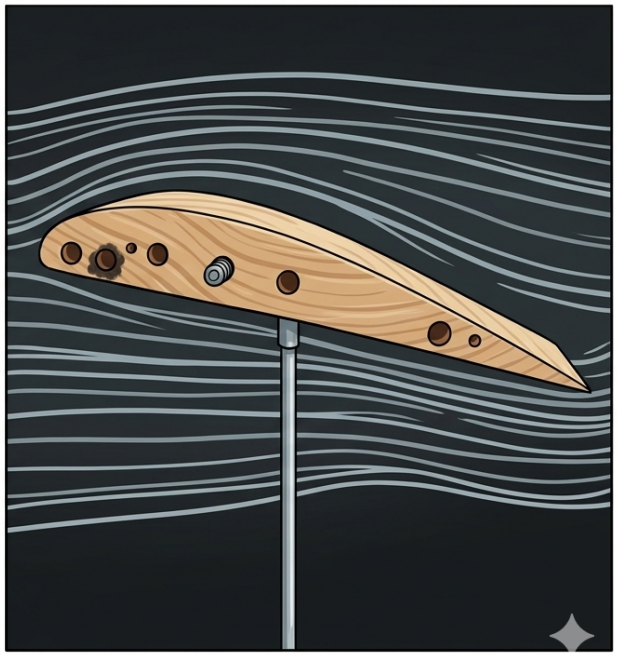
Scientific Models

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Models

- Two models of a plane
  - the model on the left is useful for training pilots
  - the model on the right is useful for engineering aerodynamics
- Models are not meant to be copies or emulations of the real thing
  - models simplify or characterize the real thing by creating an abstraction of it
  - models are not "right," but can be useful



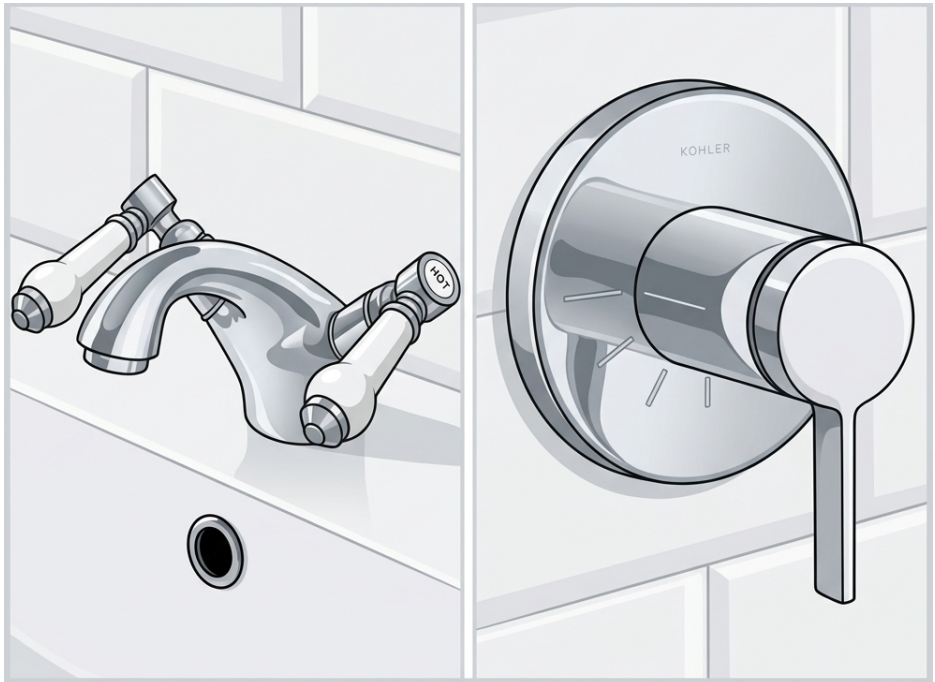


Goals of Cognitive Models

- Cognitive models focus on the cognitive phenomena of how people think and behave
  - perception, memory, learning, decision making, ...
- There are three goals models should aim for to be useful
  - description: what do the outputs look like?
  - explanation: why do they look like they do?
  - prediction: what would they look like under new or different circumstances?

Elements of Faucets

- A faucet with taps that produces water has
  - settings: the tap(s) that control the temperature and amount of flow of water
  - processes: inner workings of the faucet that produces the water temperature and flow
  - behavior: the resultant water temperature and flow that is observed and can be measured.

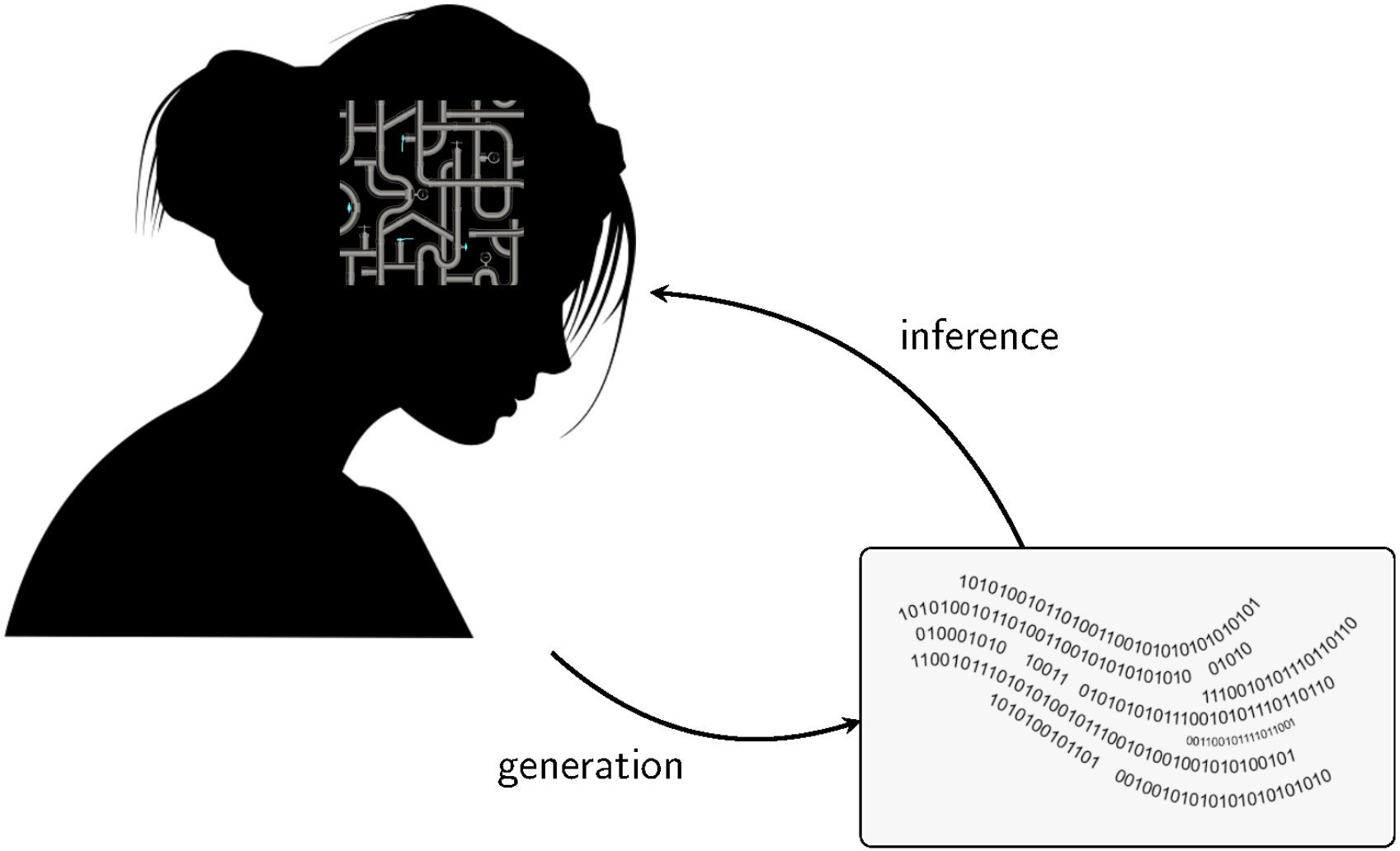


Elements of Cognitive Models

- Models in cognitive science have
  - parameters: latent (unobservable, hidden) psychological variables, which have settings that control psychological processes (e.g., working memory capacity, learning rate, IQ, . . .)
  - processes: latent psychological mechanisms that detect and comprehend stimuli, encode, process, and retrieve information, and make decisions and produce actions
  - data: observable and measurable behavior resulting from the processes (e.g., button presses, response time, verbal answers, . . .)

Generation and Inference

- Cognitive models make assumptions about how data are generated
  - this allows statistical inferences from the data about cognition



Scientific Inferences

- Statistical methods connect the modeling assumptions and the observed data to make inferences about cognition and predictions about data:
  - inference about model adequacy: is the model capable of generating the data that were observed?
  - model comparisons: which two or more competing models are more likely to have generated the data?
  - inferences about parameters: what values of the parameters are likely to have generated the observed data?
  - predictions about data: an adequate model can generate data that make predictions about observed behavior